Practical 3: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind Problem

Definition Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

CODE:-

```
<!DOCTYPE html>
```

1. index.html

```
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">

   <title>Student Github - Home</title>

   <style>
      * {
         margin: 0;
         padding: 0;
         box-sizing: border-box;
      }
```

```
body {
    font-family: Arial, sans-serif;
    background-color: #f4f6f8;
    color: #222;
}

header {
    background-color: #1e3a8a;
    color: white;
    padding: 20px;
}

header h1 {
    text-align: center;
    margin-bottom: 15px;
}

nav {
    display: flex;
    flex-direction: column;
```

```
    align-items: center;
    gap: 10px;
}

nav a {
    color: white;
    text-decoration: none;
    padding: 8px 12px;
}

nav a:hover {
    background-color: white;
    color: #1e3a8a;
    border-radius: 5px;
}

main {
    width: 90%;
    max-width: 1000px;
    margin: 40px auto;
    background-color: white;
```

```
    padding: 40px 20px;
    text-align: center;
    border-radius: 10px;
}

main h2 {
    color: #1e3a8a;
    margin-bottom: 15px;
}

main p {
    line-height: 1.6;
}

footer {
    background-color: #1e3a8a;
    color: white;
    text-align: center;
    padding: 20px;
    margin-top: 50px;
}
```

```
@media (min-width: 600px) {
    nav {
        flex-direction: row;
        justify-content: center;
        flex-wrap: wrap;
    }
}

@media (min-width: 900px) {
    header h1 {
        display: inline-block;
        margin: 0;
    }

    nav {
        float: right;
        margin-top: 5px;
    }

    main {
```

```
            min-height: 300px;
            display: flex;
            flex-direction: column;
            justify-content: center;
        }
    }
  </style>
</head>

<body>

<header>
  <h1>Student Github</h1>

  <nav>
    <a href="index.html">Home</a>
    <a href="about.html">About</a>
    <a href="register.html">Register</a>
    <a href="dashboard.html">Dashboard</a>
    <a href="event.html">Events</a>
  </nav>
```

```
</header>

<main>
  <h2>Welcome to Student Github</h2>

  <p>
    This website helps students manage profiles,
    courses, events, and feedback.
  </p>
</main>

<footer>
  <p>&copy; 2026 Student Github. All rights reserved.</p>
  <p>Email: studentgithub@gmail.com</p>
  <p>Phone: +91 12345 67890</p>
</footer>

</body>
</html>
```

About.html

```
<!DOCTYPE html>
```

```
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

    <title>Student Github - About</title>

    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            color: #222;
        }
```

```
header {
    background-color: #1e3a8a;
    color: white;
    padding: 20px;
}

header h1 {
    text-align: center;
    margin-bottom: 15px;
}

nav {
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 10px;
}

nav a {
    color: white;
    text-decoration: none;
```

```
    padding: 8px 12px;
}

nav a:hover {
    background-color: white;
    color: #1e3a8a;
    border-radius: 5px;
}

main {
    width: 90%;
    max-width: 1100px;
    margin: 40px auto;
}

h2 {
    text-align: center;
    color: #1e3a8a;
    margin-bottom: 30px;
}
```

```
.about-grid {
    display: grid;
    grid-template-columns: 1fr;
    gap: 20px;
}

.card {
    background-color: white;
    padding: 25px;
    border-radius: 10px;
}

.card h3 {
    color: #1e3a8a;
    margin-bottom: 10px;
}

.card p {
    line-height: 1.6;
}
```

```
footer {
    background-color: #1e3a8a;
    color: white;
    text-align: center;
    padding: 20px;
    margin-top: 50px;
}

@media (min-width: 600px) {
    nav {
        flex-direction: row;
        justify-content: center;
        flex-wrap: wrap;
    }

    .about-grid {
        grid-template-columns: repeat(2, 1fr);
    }
}

@media (min-width: 900px) {
```

```
        header h1 {
            display: inline-block;
            margin: 0;
        }

        nav {
            float: right;
            margin-top: 5px;
        }

        .about-grid {
            grid-template-columns: repeat(3, 1fr);
        }
    }
  </style>
</head>

<body>

<header>
  <h1>Student Github</h1>
```

```
    <nav>
        <a href="index.html">Home</a>
        <a href="about.html">About</a>
        <a href="register.html">Register</a>
        <a href="dashboard.html">Dashboard</a>
        <a href="event.html">Events</a>
    </nav>
</header>

<main>

    <h2>About Student Github</h2>

    <div class="about-grid">

        <div class="card">
            <h3>Our Platform</h3>
            <p>
                Student Github is a platform designed for
                students to manage their academic activities.
```

```
        </p>
    </div>

    <div class="card">
        <h3>Our Goal</h3>
        <p>
            Our goal is to provide students with a simple
            and responsive website.
        </p>
    </div>

    <div class="card">
        <h3>Technology</h3>
        <p>
            This website is developed using HTML and CSS
            with responsive design.
        </p>
    </div>

</div>
```

```
</main>

<footer>
    <p>&copy; 2026 Student Github. All rights reserved.</p>
</footer>

</body>
</html>
```

3. register.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Student Github - Registration</title>

    <style>
        * {
            margin: 0;
```

```
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    background-color: #f4f6f8;
    color: #222;
}

header {
    background-color: #1e3a8a;
    color: white;
    padding: 20px;
}

header h1 {
    text-align: center;
    margin-bottom: 15px;
}
```

```
nav {
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 10px;
}

nav a {
    color: white;
    text-decoration: none;
    padding: 8px 12px;
}

nav a:hover {
    background-color: white;
    color: #1e3a8a;
    border-radius: 5px;
}

main {
    width: 90%;
```

```
    max-width: 600px;
    margin: 40px auto;
}

h2 {
    text-align: center;
    color: #1e3a8a;
    margin-bottom: 25px;
}

form {
    background-color: white;
    padding: 25px;
    border-radius: 10px;
}

.form-group {
    display: flex;
    flex-direction: column;
    margin-bottom: 15px;
}
```

```
label {
    margin-bottom: 5px;
    font-weight: bold;
}

input,
select {
    padding: 12px;
    border: 1px solid #ccc;
    border-radius: 5px;
}

button {
    width: 100%;
    padding: 12px;
    background-color: #1e3a8a;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
```

```
}

button:hover {
    background-color: #162d6b;
}

footer {
    background-color: #1e3a8a;
    color: white;
    text-align: center;
    padding: 20px;
    margin-top: 50px;
}

@media (min-width: 600px) {
    nav {
        flex-direction: row;
        justify-content: center;
        flex-wrap: wrap;
    }
```

```
        form {
            padding: 35px;
        }
    }

    @media (min-width: 900px) {
        header h1 {
            display: inline-block;
            margin: 0;
        }

        nav {
            float: right;
            margin-top: 5px;
        }
    }
  </style>
</head>

<body>
```

```
<header>
   <h1>Student Github</h1>

   <nav>
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="register.html">Register</a>
      <a href="dashboard.html">Dashboard</a>
      <a href="event.html">Events</a>
   </nav>
</header>

<main>

   <h2>Student Registration</h2>

   <form>

      <div class="form-group">
         <label>Name</label>
         <input type="text" placeholder="Enter your name">
```

```
    </div>

    <div class="form-group">
      <label>Email</label>
      <input type="email" placeholder="Enter your email">
    </div>

    <div class="form-group">
      <label>Username</label>
      <input type="text" placeholder="Enter username">
    </div>

    <div class="form-group">
      <label>Password</label>
      <input type="password" placeholder="Enter
password">
    </div>

    <div class="form-group">
      <label>Course</label>
```

```
        <select>
          <option>Select Course</option>
          <option>B.Tech CE</option>
          <option>B.Tech CSE</option>
          <option>B.Tech IT</option>
        </select>
      </div>

      <button type="submit">Register</button>

    </form>

</main>

<footer>
    <p>&copy; 2026 Student Github. All rights reserved.</p>
</footer>

</body>
</html>
```

Dashboard html:-

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

    <title>Student Github - Dashboard</title>

    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            color: #222;
        }
```

```
header {
    background-color: #1e3a8a;
    color: white;
    padding: 20px;
}

header h1 {
    text-align: center;
    margin-bottom: 15px;
}

nav {
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 10px;
}

nav a {
    color: white;
```

```
    text-decoration: none;
    padding: 8px 12px;
}

nav a:hover {
    background-color: white;
    color: #1e3a8a;
    border-radius: 5px;
}

main {
    width: 90%;
    max-width: 1100px;
    margin: 40px auto;
}

h2 {
    text-align: center;
    color: #1e3a8a;
    margin-bottom: 30px;
}
```

```
.dashboard-grid {
    display: grid;
    grid-template-columns: 1fr;
    gap: 20px;
}

.card {
    background-color: white;
    padding: 30px;
    text-align: center;
    border-radius: 10px;
}

.card h3 {
    color: #1e3a8a;
    font-size: 30px;
    margin-bottom: 10px;
}

footer {
```

```
    background-color: #1e3a8a;
    color: white;
    text-align: center;
    padding: 20px;
    margin-top: 50px;
}

@media (min-width: 600px) {
    nav {
        flex-direction: row;
        justify-content: center;
        flex-wrap: wrap;
    }

    .dashboard-grid {
        grid-template-columns: repeat(2, 1fr);
    }
}

@media (min-width: 900px) {
    header h1 {
```

```
            display: inline-block;
            margin: 0;
        }

        nav {
            float: right;
            margin-top: 5px;
        }

        .dashboard-grid {
            grid-template-columns: repeat(3, 1fr);
        }
    }
  </style>
</head>

<body>

<header>
  <h1>Student Github</h1>
```

```
    <nav>
        <a href="index.html">Home</a>
        <a href="about.html">About</a>
        <a href="register.html">Register</a>
        <a href="dashboard.html">Dashboard</a>
        <a href="event.html">Events</a>
    </nav>
</header>

<main>

    <h2>Student Dashboard</h2>

    <div class="dashboard-grid">

        <div class="card">
            <h3>120</h3>
            <p>Total Students</p>
        </div>

        <div class="card">
```

```
            <h3>15</h3>
            <p>Total Events</p>
        </div>

        <div class="card">
            <h3>85</h3>
            <p>Total Registrations</p>
        </div>

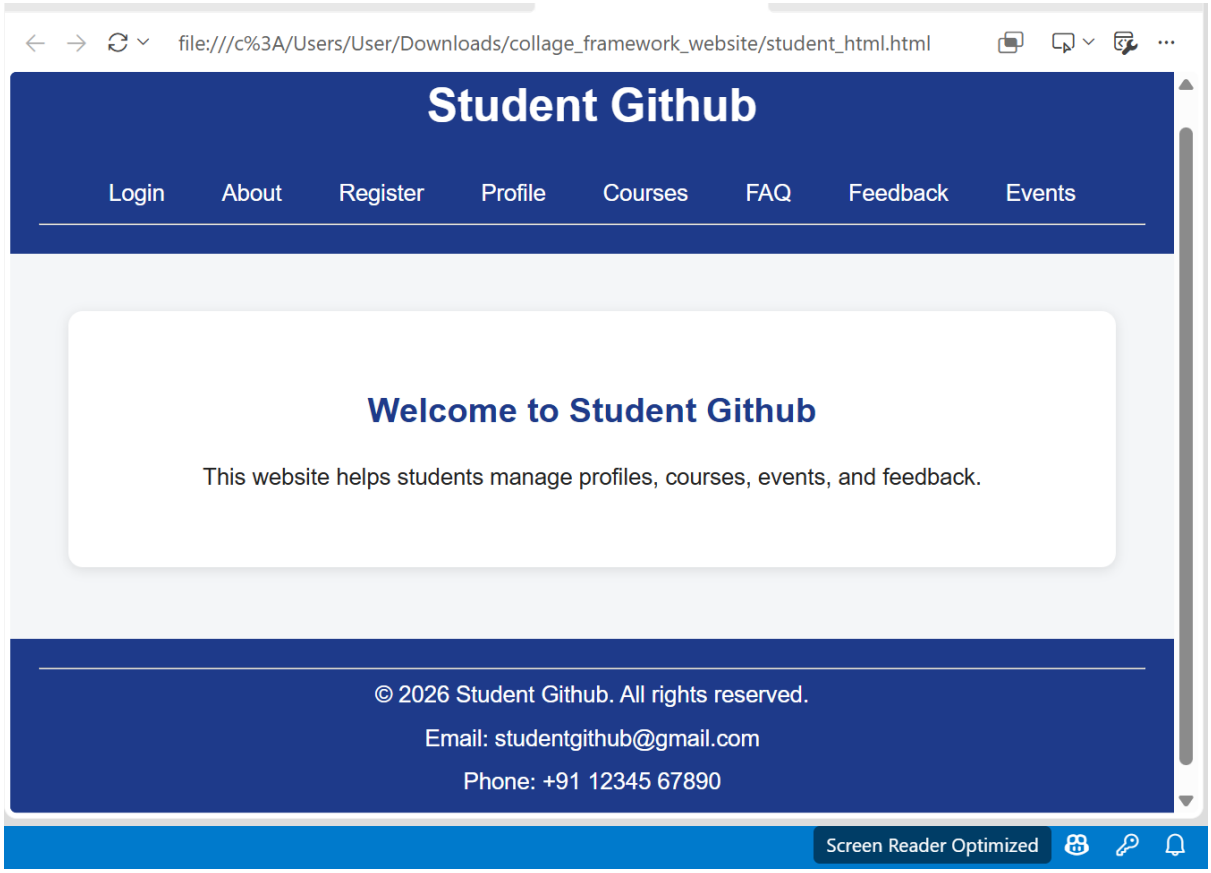
    </div>

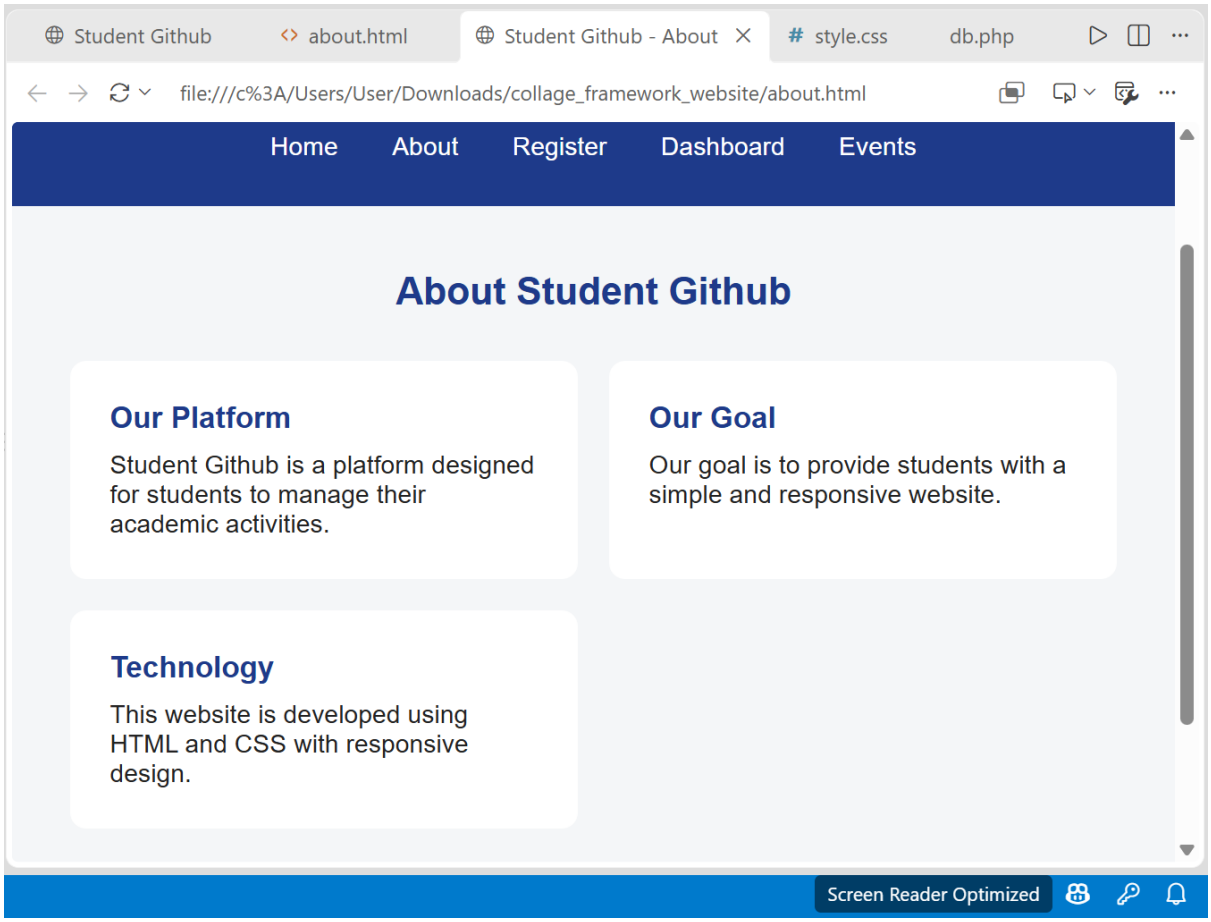
</main>

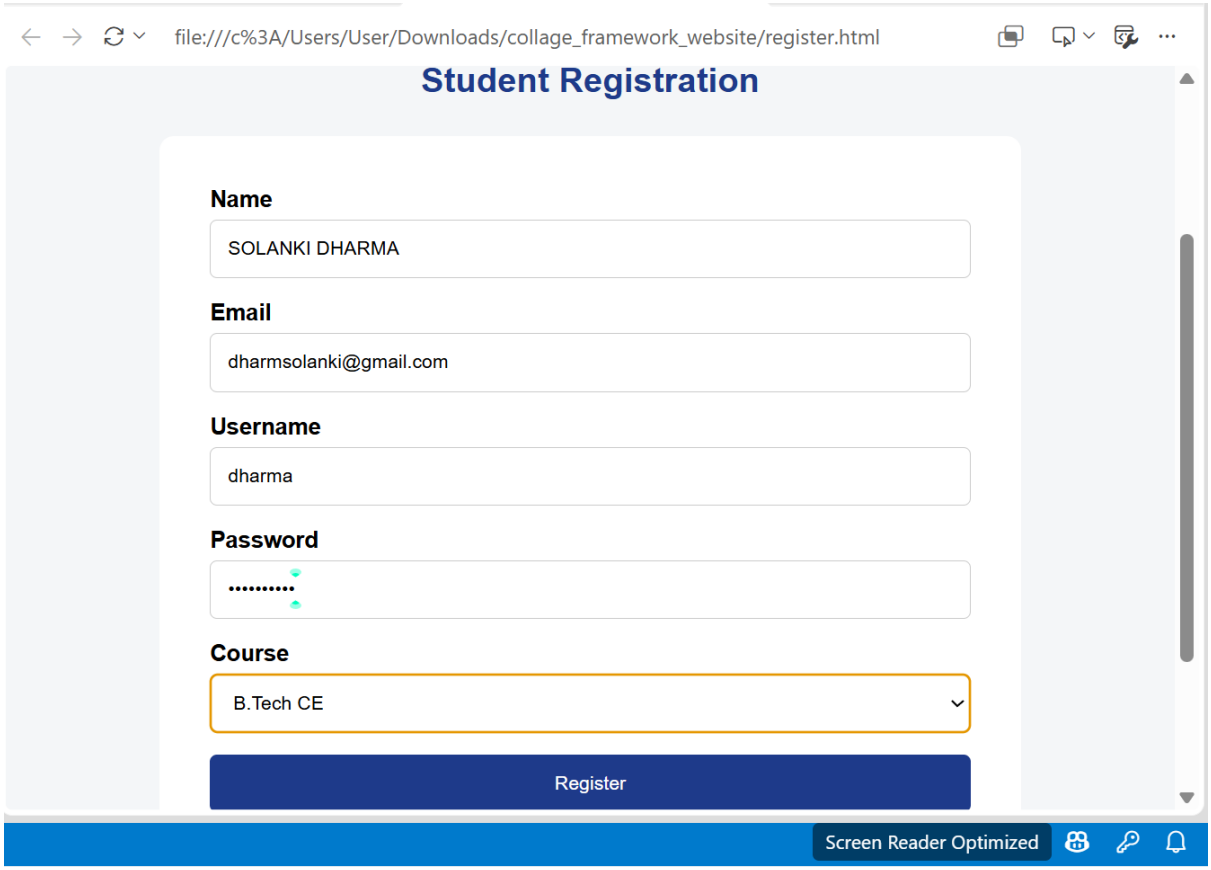
<footer>
    <p>&copy; 2026 Student Github. All rights reserved.</p>
</footer>

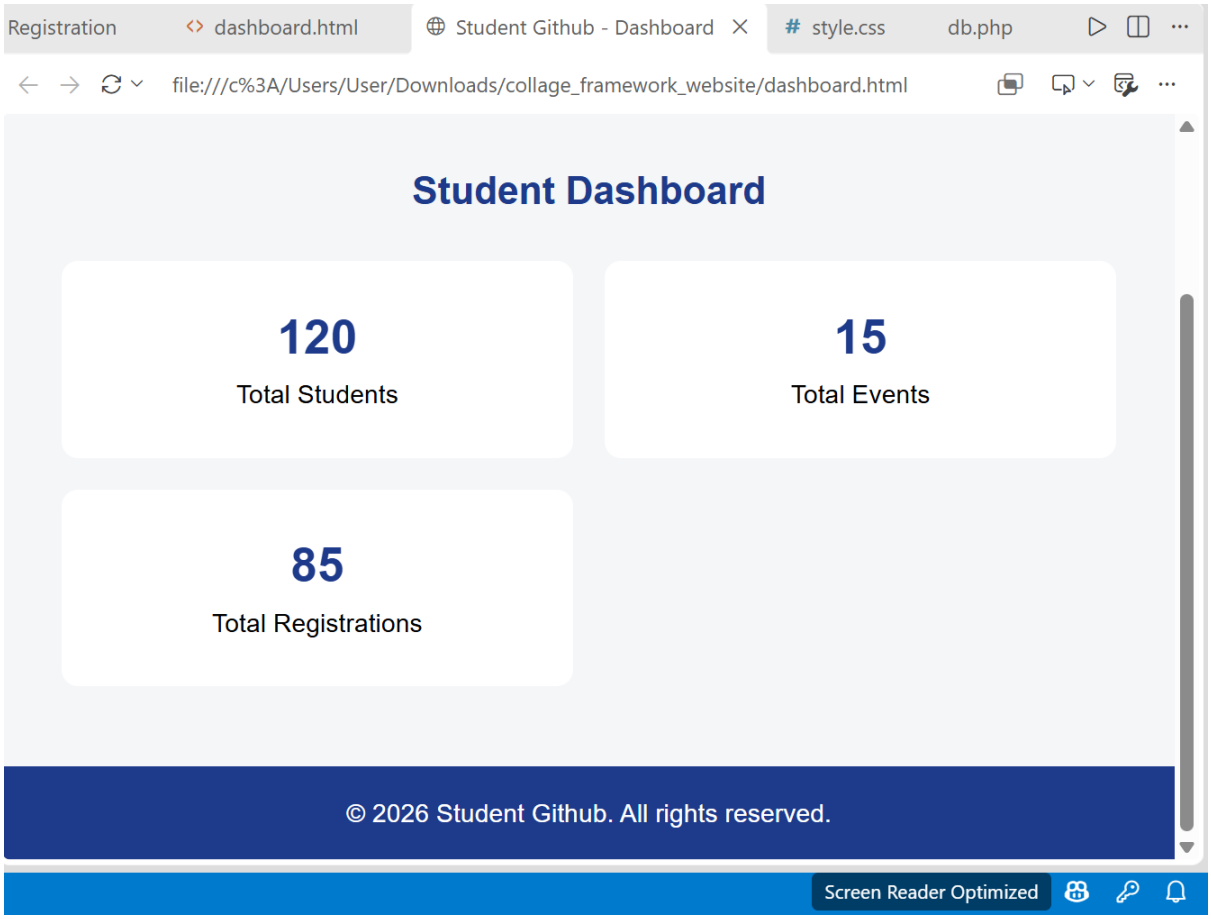
</body>
</html>
```

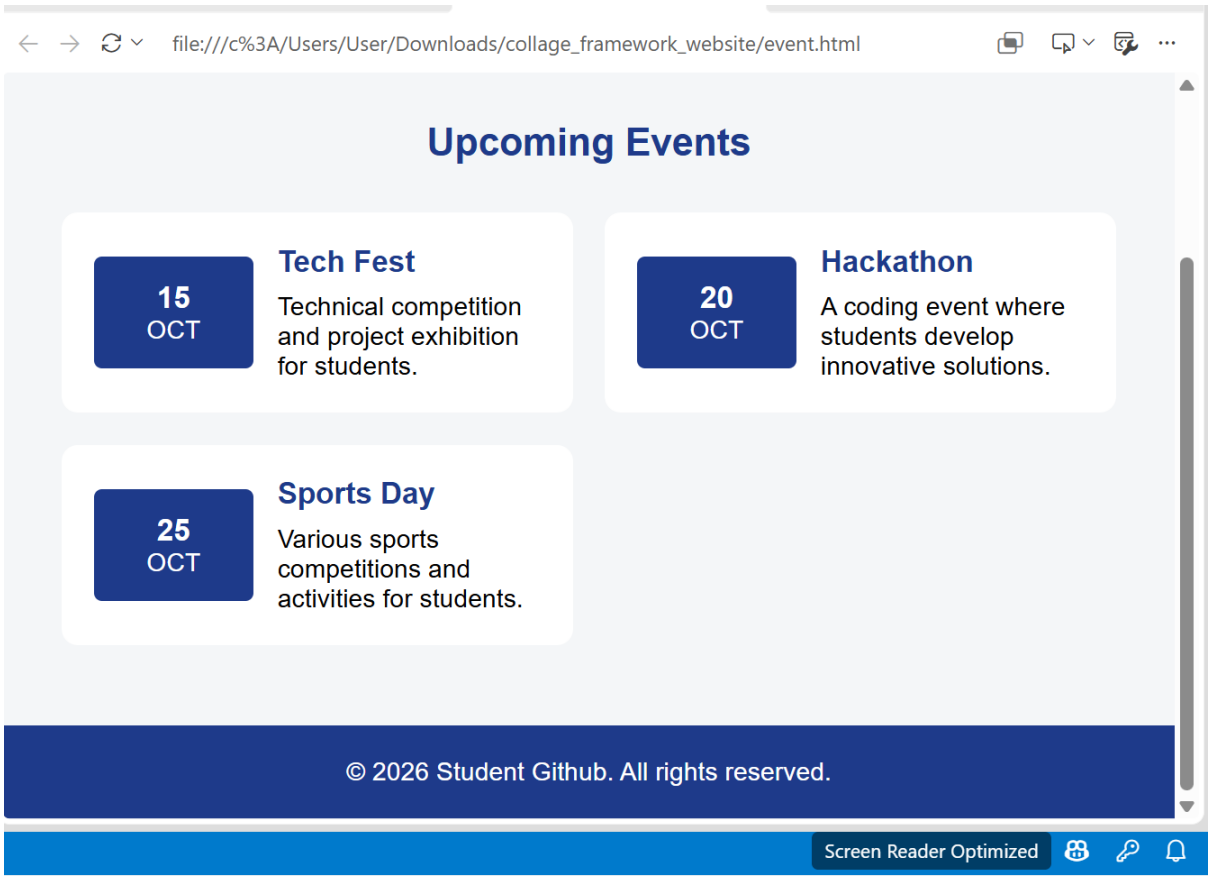
OUTPUT:-


file:///c%3A/Users/User/Downloads/collage_framework_website/student_html.html
Student Github
Login
About
Register
Profile
Courses
FAQ
Feedback
Events
Welcome to Student Github
This website helps students manage profiles, courses, events, and feedback.
© 2026 Student Github. All rights reserved.
Email: studentgithub@gmail.com
Phone: +91 12345 67890
Screen Reader Optimized


Student Github
about.html
Student Github - About
style.css
db.php
file:///c%3A/Users/User/Downloads/collage_framework_website/about.html
Home
About
Register
Dashboard
Events
About Student Github
Our Platform
Student Github is a platform designed for students to manage their academic activities.
Our Goal
Our goal is to provide students with a simple and responsive website.
Technology
This website is developed using HTML and CSS with responsive design.
Screen Reader Optimized


file:///c%3A/Users/User/Downloads/collage_framework_website/register.html
Student Registration
Name
SOLANKI DHARMA
Email
dharmsolanki@gmail.com
Username
dharma
Password
Course
B.Tech CE
Register
Screen Reader Optimized


Registration
dashboard.html
Student Github - Dashboard
style.css
db.php
file:///c%3A/Users/User/Downloads/collage_framework_website/dashboard.html
Student Dashboard
120
Total Students
15
Total Events
85
Total Registrations
© 2026 Student Github. All rights reserved.
Screen Reader Optimized


file:///c%3A/Users/User/Downloads/collage_framework_website/event.html
Upcoming Events
15
OCT
Tech Fest
Technical competition and project exhibition for students.
20
OCT
Hackathon
A coding event where students develop innovative solutions.
25
OCT
Sports Day
Various sports competitions and activities for students.
© 2026 Student Github. All rights reserved.
Screen Reader Optimized

Key Questions / Analysis / Interpretation

1. How does the layout adapt for mobile, tablet, and desktop screens?

The website uses responsive CSS media queries. On mobile, elements are arranged in a single column. On tablet and desktop, Flexbox and CSS Grid change the layout into multiple columns and rows according to screen size.

2. Which layout technique is used and why?

CSS Flexbox and CSS Grid are used. Flexbox is mainly used for navigation and arranging items in a row or column. CSS Grid is used for cards, events, and dashboard sections. They make the layout flexible and responsive.

3. Are typography, spacing, colors, and contrast readable?

Yes. The website uses a simple Arial font, proper spacing, readable font sizes, and good contrast between the dark blue background and white text. This makes the content easy to read.

4. Is the framework used consistently without unnecessary inline CSS?

Yes. No external framework such as Bootstrap or

Tailwind is used. CSS is written inside the <style> section of each HTML file, keeping the styling organized and avoiding unnecessary inline style="" attributes.

Tools/Technology to be Used:-

VS code HTML5, CSS3, Bootstrap, browser responsive mode